

Theoretical prediction of breakthrough curves for molecular sieve adsorption columns—I Asymptotic solutions

D. R. GARG and D. M. RUTHVEN

Department of Chemical Engineering, University of New Brunswick, Fredericton, N.B., Canada

(Received 11 April 1972; accepted 18 July 1972)

Abstract—The asymptotic forms of the breakthrough curves for molecular sieve adsorption columns are described. For column saturation, the limiting breakthrough curve is of the 'constant pattern' type. For the two cases of micropore and macropore diffusion control the forms of the constant pattern curves are calculated from the solutions of the appropriate diffusion equations subject to the simplified boundary conditions appropriate for a long column. The analysis, which is restricted to the adsorption of a single adsorbable component from a lean gas, should provide an appropriate approximate description of the limiting behaviour of a molecular sieve column when operated under isothermal conditions. The 'proportionate pattern' type of asymptotic breakthrough curve which is obtained for regeneration of a saturated column is briefly discussed.

INTRODUCTION

THE PRACTICAL applications of molecular sieves as drying agents and selective adsorbents generally involve the use of packed adsorption columns. For the proper design of such processes knowledge of the breakthrough characteristics of the column is required. The problem of predicting breakthrough curves from basic kinetic and equilibrium data has attracted much attention because of its importance not only in connection with adsorption columns but also in relation to chromatography and ion exchange. In principle the breakthrough curve may be calculated, for any system of known kinetics, from the solution of the differential rate equation subject to the boundary conditions imposed by the differential fluid phase mass balance for an element of the column. In practice the differential equations are generally complex and full solutions have been obtained only for the simpler cases. Useful insight may however be obtained from the asymptotic solutions which give the limiting forms of the breakthrough curves for sufficiently long columns. Such solutions are relatively easy to obtain since, in the asymptotic limit, the boundary conditions imposed by the mass balance assume a simplified form. A useful

general review covering both general and asymptotic solutions has been given by Vermeulen[1]. The present paper is concerned only with those solutions which provide appropriate descriptions of the asymptotic behaviour of molecular sieve columns.

KINETICS AND EQUILIBRIUM OF SORPTION

The equilibrium isotherms for the occlusion of many species in molecular sieve zeolites are of the Langmuir form. Such isotherms may be classified as 'favorable' for adsorption and 'unfavourable' for desorption. The physical conditions of the idealized Langmuir model are generally not fulfilled but the Langmuir equation nevertheless provides a useful empirical correlation:

$$\frac{q^*}{q_s} = \frac{bc}{1 + bc} \quad (1)$$

The parameters q_s and b should however be regarded simply as empirical constants[2].

Commercial molecular sieves consist of small crystals of zeolite (0.5–5.0 μ) pelleted with a clay binder. Such solids therefore have a bidisperse pore structure and the kinetics of ad-

sorption or desorption are governed by three distinct diffusional resistances:

- (i) film diffusion or external mass transfer from the bulk fluid to the pellet surface;
- (ii) diffusion through the macropores of the pellet;
- (iii) diffusion within the micropores of the zeolite crystals.

Under conditions of practical importance film resistance is generally negligible and the rate of sorption is controlled by the macropore and micropore resistances. Under equilibrium conditions and for sorbates which are occluded by the zeolite, the quantity of sorbate held within the zeolite crystals is generally far greater than the quantity remaining within the macropores of the pellet. In the analysis of the kinetics of sorption, uptake by the macropores is therefore neglected.

The relative importance of macropore and micropore diffusional resistance depends on the particular system and on the conditions and may be estimated from the magnitude of the

$$\text{parameter } \Omega = \frac{3w(1-\epsilon_p)}{\epsilon_p} \frac{D_z}{D_p} \left[\frac{R_p}{r_z} \right]^2 \frac{dq^*}{dc} [3]$$

($\Omega < \sim 1$, micropore control; $\Omega > \sim 100$, macropore control). For a general treatment of the sorption kinetics it is in principle necessary to take account of both micropore and macropore resistances. It is however convenient to consider only the limiting cases in which one or other resistance is dominant. Such an analysis is directly applicable to many real systems and serves to illustrate the principal features of column performance.

For the case of micropore diffusion control we assume that transport within the macropores of the pellet is sufficiently rapid to maintain a uniform sorbate concentration $\bar{c}$ equal to the bulk phase concentration c throughout the macropores of any particular pellet. Sorption at the crystal surface is assumed to be rapid so that the surface concentration of sorbate is in equilibrium with the bulk fluid at the particular point in the column. Assuming that the zeolite crystals can

be treated as uniform spherical particles the appropriate form of the diffusion equation is:

$$\frac{\partial q}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_z r^2 \frac{\partial q}{\partial r} \right). \quad (2)$$

If the zeolitic diffusivity D_z is independent of sorbate concentration, Eq. (2) reduces to the familiar Fickian expression:

$$\frac{\partial q}{\partial t} = D_z \cdot \left(\frac{\partial^2 q}{\partial r^2} + \frac{2}{r} \cdot \frac{\partial q}{\partial r} \right). \quad (3)$$

In general, however, zeolitic diffusivities are concentration dependent. For the diffusion of water[4] and light hydrocarbons in several zeolites[5, 6] the concentration dependence of the diffusivity has been shown to be related to the shape of the equilibrium isotherm by the Darken equation:

$$D_z = D_* \frac{\partial \ln a}{\partial \ln q}. \quad (4)$$

When the equilibrium isotherm obeys the Langmuir equation this relationship becomes:

$$D_z = \frac{D_*}{1 - q/q_s} \quad (5)$$

where D_* , the limiting diffusivity at zero concentration, is constant.

The corresponding (non-linear) form of the diffusion equation is:

$$\frac{\partial q}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{D_* r^2}{1 - q/q_s} \cdot \frac{\partial q}{\partial r} \right) \quad (6)$$

and the appropriate boundary conditions are:

$$q(r_z, t - z/v) = q^* = \frac{q_s bc}{1 + bc}; \quad \frac{\partial q}{\partial r} (0, t - z/v) = 0 \quad (7)$$

with the initial conditions, for a step change in concentration at the column inlet at time zero being:

$$q(r, 0) = 0 \quad (\text{saturation}) \quad (8)$$

$$q(r, 0) = q_0 \quad (\text{regeneration}). \quad (9)$$

$$\bar{q}(R, 0) = 0 \quad (\text{saturation}) \quad (15)$$

Since the sorbate is assumed to be held entirely by the zeolite crystals it is convenient to express the adsorbed phase concentration on the basis of unit zeolite crystal volume:

$$\bar{q} = \bar{q} = \frac{3}{r_z^3} \int_0^{r_z} q r^2 \cdot dr. \quad (10)$$

$$\bar{q}(R, 0) = q_0 \quad (\text{regeneration}) \quad (16)$$

and the average solid phase concentration at any point in the column is:

$$\bar{q} = \frac{3}{R_p^3} \int_0^{R_p} \bar{q} R^2 dR. \quad (17)$$

For macropore control it is assumed that diffusion within the zeolite crystals is sufficiently rapid to maintain at all times a uniform sorbate concentration through any particular crystal. Rapid equilibration at the surface is assumed so that the sorbed fluid is in equilibrium with the macropore fluid just outside the crystal (i.e. $q = \bar{q} = q^*$). For a system of uniform spherical pellets the relevant form of the diffusion equation, neglecting gas phase accumulation in the macropores, is:

$$\frac{\partial \bar{q}}{\partial t} = \frac{\epsilon_p}{w(1-\epsilon_p)} \frac{D_p}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial \bar{q}}{\partial R} \right) \quad (11)$$

For a Langmuir isotherm (Eq. 1):

$$\frac{\partial \bar{q}}{\partial \bar{c}} = \frac{(1+b\bar{c})^2}{bq_s} = \frac{1}{bq_s} \cdot \frac{1}{(1-\bar{q}/q_s)^2} \quad (12)$$

For most systems the macropore diffusivity is essentially independent of sorbate concentration so that the diffusion equation becomes:

$$\frac{\partial \bar{q}}{\partial t} = \frac{\epsilon_p}{w(1-\epsilon_p)} \frac{D_p}{bq_s} \cdot \frac{1}{R^2} \frac{\partial}{\partial R} \left[\frac{R^2}{(1-\bar{q}/q_s)^2} \cdot \frac{\partial \bar{q}}{\partial R} \right] \quad (13)$$

This equation is formally the same as the particle diffusion equation with a concentration dependent effective diffusivity given by

$$D_{ep} = \frac{\epsilon_p D_p}{w(1-\epsilon_p) bq_s (1-\bar{q}/q_s)^2}$$

boundary and initial conditions are:

$$\bar{q}(R_p, t-z/v) = \frac{q_s bc}{1+bc}; \quad \frac{\partial \bar{q}}{\partial R}(0, t-z/v) = 0 \quad (14)$$

The differential fluid phase mass balance, which relates the average sorbate concentration in the solid to the concentration in the bulk fluid at any point in the column completes the system of equations defining the breakthrough curve.

FLUID PHASE MASS BALANCE

The present analysis is restricted to isothermal systems involving dilute fluid mixtures containing only one adsorbable component. Pressure drop across the adsorbent bed is neglected and plug flow is assumed. The fluid phase mass balance for an element of the bed at a distance z from the column inlet is given by:

$$v \cdot \frac{\partial c}{\partial z} + \frac{\partial c}{\partial t} + \frac{1}{m} \cdot \frac{\partial \bar{q}}{\partial t} = 0 \quad (18)$$

where $m = \epsilon/(1-\epsilon') =$ ratio of bed void volume to crystal volume. For saturation of an initially empty column the appropriate initial and boundary conditions are:

$$\bar{q}(z, 0) = 0; \quad c(0, t) = c_0 \quad (19)$$

while the corresponding conditions for regeneration of a saturated column are:

$$\bar{q}(z, 0) = q_0; \quad c(0, t) = 0. \quad (20)$$

To obtain the complete solution for the breakthrough curve for micropore diffusion control it is necessary to solve Eq. (6) subject to the boundary conditions defined by Eqs. (7)-(10) and (18)-(20). The corresponding problem for macropore control requires the solution of Eq. (13) subject to the boundary conditions (14)-

(20). Since the equations are non-linear, numerical methods offer the only feasible approach. The numerical calculations are however bulky and require considerable computational time. It is therefore appropriate to consider, as a first approach to the problem, the asymptotic solutions.

NATURE OF THE ASYMPTOTIC SOLUTIONS

The nature of the asymptotic form of the breakthrough curve depends on the equilibrium isotherm. When the isotherm is unfavourable, as for the regeneration of a saturated column with a non-adsorbing eluent, the concentration front broadens as it progresses through the column. For a sufficiently long column or a sufficiently low flowrate, equilibrium is essentially reached and Eq. (18) may be integrated at constant c to give, after some manipulation:

$$\frac{vt}{z} - 1 = \frac{1}{m} \cdot \frac{d\bar{q}}{dc} \quad (21)$$

The shape of the asymptotic breakthrough curve is determined only by the equilibrium isotherm and is independent of the kinetics of sorption. For a Langmuir isotherm we have, from Eqs. (12) and (21):

$$\frac{vt}{z} - 1 = \frac{bq_s}{m} \cdot \frac{1}{(1 + bc)^2} \quad (22)$$

or, expressed in terms of the dimensionless variables X/T and the parameter λ :

$$\frac{c}{c_0} = \left(\frac{1 - \lambda}{\lambda} \right) \left(\sqrt{\frac{X}{T}} - 1 \right) \quad (23)$$

$$(1 < X/T < (1 - \lambda)^{-2}).$$

The asymptotic form of the breakthrough curve is thus of the 'proportionate pattern' type since, for a given system, the concentration is a function only of the single variable X/T or vt/z .

The argument given above, which is due to De Vault[7], gives no information concerning the length of column required to approach the asymptotic solution. Such information can be obtained only from the full solution of the prob-

lem. Thus, although Eq. (23) gives, for any specified conditions, a simple and practically useful approximate prediction of the time required to regenerate a saturated column, the validity of the approximation can be determined only by a more complete analysis in which the kinetics of desorption are considered.

When the isotherm is favourable, as for the adsorption of most species in molecular sieves, the asymptotic behaviour of the column is quite different. The physical reason for this difference has been discussed by Lightfoot *et al.*[8]. The rate at which a sorbate travels along the column depends on the product of the fluid velocity and the sorbate concentration in the fluid phase. For a favourable isotherm the ratio of fluid phase to adsorbed phase concentrations increases with concentration, so that the sorbate moves more rapidly when the concentration is high. The concentration front is thus self-sharpening and the asymptotic limit is reached when the concentration profiles in fluid and solid phases become coincident. Under these conditions the fluid phase mass balance reduces to the simple equation:

$$c/c_0 = \bar{q}/q_0. \quad (24)$$

The shape of the concentration profile becomes independent of column lengths and the front travels with a uniform velocity giving rise to a 'constant pattern' breakthrough curve. In contrast with the 'proportionate pattern' case the shape of the asymptotic breakthrough curve is determined by both the kinetics and equilibrium of sorption. For sufficiently rapid sorption a step function will be approached.

CONSTANT PATTERN BREAKTHROUGH CURVES FOR COLUMN SATURATION

To calculate the form of the asymptotic constant pattern breakthrough curve for column saturation requires the solution of Eqs. (6) or (13) but with the boundary conditions defined by Eqs. (18) and (19) replaced by the much simpler condition expressed by Eq. (24). In order to generalize the numerical analysis it is

convenient to express the equations in dimensionless form. With the dimensionless variables and parameters: $\phi = c/c_0$, $\psi = q/q_0$, $\lambda = q_0/q_s = bc_0(1+bc_0)^{-1}$, $T = (D_*/r_z^2)(t-z/v)$, $D_1 = D_z/D_*$, $D_1/D_* = (1-\lambda\psi)^{-1}$, $\eta = r/r_z$,

$$s = \frac{\int_0^\psi D_1 \cdot d\psi}{\int_0^1 D_1 \cdot d\psi} = \frac{\ln(1-\lambda\psi)}{\ln(1-\lambda)},$$

$$u = s\eta, \beta = -\ln(1-\lambda)$$

Eq. (6), the rate equation for micropore diffusion control, reduces to:

$$\frac{\partial u}{\partial T} = e^{\beta u/\eta} \cdot \frac{\partial^2 u}{\partial \eta^2}. \quad (25)$$

The modified initial and boundary conditions are:

$$u(\eta, 0) = 0 \quad (26)$$

$$u(1, T) = -\frac{1}{\beta} \ln \left[1 - \frac{\lambda\phi(1+bc_0)}{1+bc_0\phi} \right]; u(0, T) = 0 \quad (27)$$

and the dimensionless average sorbate concentration in the solid is given by:

$$\bar{\psi} = \bar{\psi} = \frac{3}{\lambda} \int_0^1 (1 - e^{-\beta u/\eta}) \eta^2 \cdot d\eta. \quad (28)$$

For the case of macropore diffusion control the appropriate dimensionless variables are:

$$\bar{\psi} = \bar{q}/q_0,$$

$$T' = \frac{\epsilon_p D_p (t-z/v)}{w(1-\epsilon_p) R_p^2 b q_s}, \eta' = R/R_p,$$

$$D'_1 = (1-\lambda\bar{\psi})^{-2},$$

$$s' = \frac{\int_0^{\bar{\psi}} D'_1 \cdot d\bar{\psi}}{\int_0^1 D'_1 \cdot d\bar{\psi}} = \bar{\psi}(1-\lambda)(1-\lambda\bar{\psi})^{-1}, u' = s'\eta'.$$

Equation (13) is transformed to:

$$\frac{\partial u'}{\partial T'} = \left\{ \frac{1-\lambda(1-u'/\eta')}{1-\lambda} \right\}^2 \cdot \frac{\partial^2 u'}{\partial \eta'^2} \quad (29)$$

and the initial and boundary conditions given by Eqs. (15) and (14) become:

$$u'(\eta', 0) = 0 \quad (30)$$

$$u'(1, T') = \frac{(1-\lambda)(\phi + bc_0\phi)}{(1+bc_0\phi) - \lambda\phi(1+bc_0)};$$

$$u'(0, T') = 0.$$

The dimensionless average sorbate concentration in the solid is:

$$\bar{\bar{\psi}} = 3 \int_0^1 \frac{u'\eta'^2 d\eta'}{\eta' - \lambda(\eta' - u')}. \quad (32)$$

Equation (24), which applies for both micropore and macropore control, becomes:

$$\phi = \bar{\bar{\psi}}. \quad (33)$$

Although the boundary conditions are different, Eqs. (25) and (29) are formally the same as the equations used in our earlier paper[9] to describe the uptake curves for a single particle of zeolite subjected to a step change in sorbate concentration, and the solutions may be obtained by similar numerical methods. An implicit finite difference scheme (Crank-Nicholson method) was used. Further details are given in the appendix to the next paper. An arbitrary initial incremental value was assigned to ϕ in order to start the solution but for all subsequent time steps the condition expressed by Eq. (33) was used. Provided that the initial increment was sufficiently small (0.002) convergence to the asymptotic solution was rapid. Some error is however to be expected at very small ϕ values ($\phi < 0.02$).

Asymptotic constant pattern breakthrough curves, calculated in the above manner for both micropore and macropore diffusion control, are shown in Figs. 1 and 2 and tabulated numerical

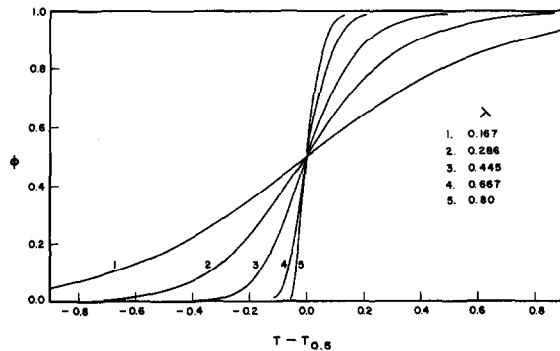


Fig. 1. Asymptotic constant pattern breakthrough curves for micropore diffusion control with concentration dependent diffusivity.

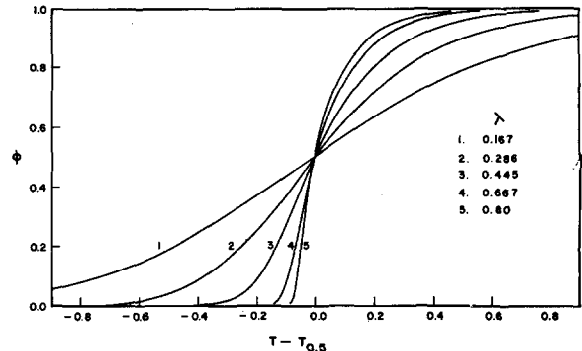


Fig. 3. Asymptotic constant pattern breakthrough curves for micropore diffusion control with constant diffusivity.

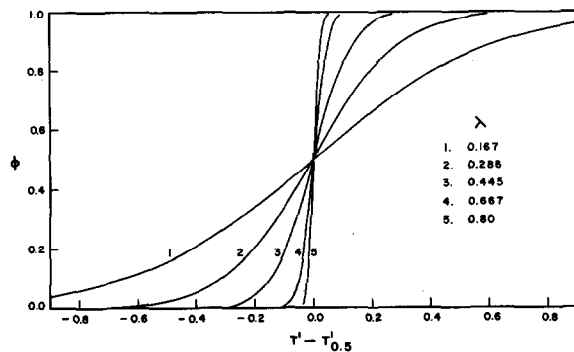


Fig. 2. Asymptotic constant pattern breakthrough curves for macropore diffusion control.

values are given in the thesis of Garg[10]. In order to illustrate the effect of the concentration dependence of the zeolitic diffusivity on the breakthrough curves, for the micropore control case, the set of curves given in Fig. 3 was calculated, for the same values of λ , assuming a constant diffusivity. For this calculation Eq. (25) was replaced by the dimensionless form of Eq. (3):

$$\frac{\partial u}{\partial T} = \frac{\partial^2 u}{\partial \eta^2}. \quad (34)$$

For the cases of macropore control and micropore control with constant diffusivity a previous numerical solution has been given by Hall *et al.* [11] who used a somewhat different method of

analysis. The curves shown in Figs. (2) and (3) appear to be consistent with Hall's solutions although exact comparison is difficult.

The breakthrough curves for $\lambda = 0.8$ and $\lambda = 0.167$ are compared on normal probability coordinates in Fig. 4. Even at the smaller values of λ the curves show significant differences from symmetric error functions and the influence of the sorption kinetics is clearly evident. At large values of λ the curves shown in Figs. 1 and 2 (concentration dependent effective diffusivities)

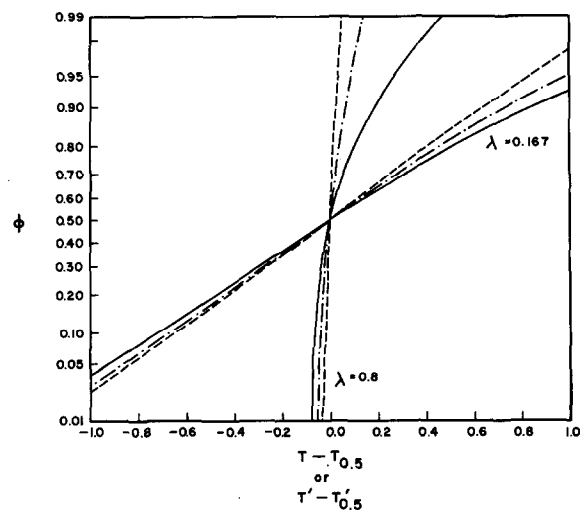


Fig. 4. Comparison of constant pattern breakthrough curves (micropore control, constant diffusivity —; micropore control, concentration dependent diffusivity ---; macropore control ----).

approach a step function which is the limiting form of the asymptotic breakthrough curve for infinitely rapid kinetics. The differences between the curves for micropore diffusion control with constant and concentration dependent diffusivities become pronounced at large λ values whereas the differences between the curves for macropore control and the corresponding curves for micropore control with a concentration dependent diffusivity are less marked. Practically, this means that to distinguish between macropore and micropore control on the basis of experimental breakthrough curves alone is quite difficult, although the shape of the breakthrough curves for micropore diffusion controlled systems should clearly reflect the effect of a concentration dependent diffusivity.

Figure 5 shows the midpoint slopes of the breakthrough curves plotted against λ . For a

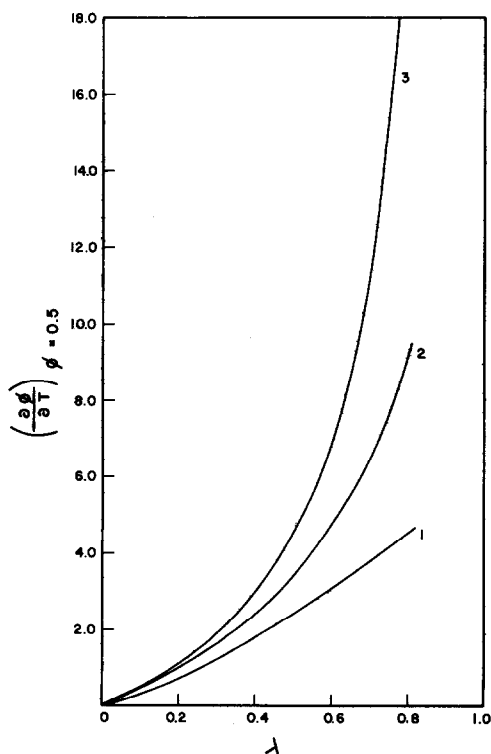


Fig. 5. Midpoint slopes of constant pattern breakthrough curves. (Curve 1—micropore control, constant diffusivity; curve 2—micropore control, concentration dependent diffusivity; curve 3—macropore control.)

system in which λ is known from the equilibrium isotherm, the midpoint slope provides a simple and convenient means of extracting the diffusional time constant (D_*/r_z^2 for micropore control or $\epsilon_p D_p / w(1 - \epsilon_p) R_p^2 b q_s$, for macropore control) from experimental breakthrough curves.

The above method of calculation yields only the shape of the asymptotic breakthrough curve: the location of the curve is determined by the length of the column and the feed flowrate. For convenience the $\phi = 0.5$ point for each curve is taken as the zero for the time scale since this facilitates matching the theoretical curves to experimental results. For a perfectly symmetrical breakthrough curve the time required to reach the midpoint will be given by $Vq_0(1 - \epsilon')/Qc_0$. For an asymmetric breakthrough curve the midpoint should strictly be defined as the point at which the areas above and below the breakthrough curve are equal, but the percentage difference in dimensionless time between this point and the point given by $\phi = 0.5$ is small when the column is sufficiently long for the asymptotic solutions to be applicable.

CONCLUSION

Although many approximations are involved in the analysis, the asymptotic breakthrough curves given in Figs. 1 and 2 and by Eq. (23) should provide appropriate descriptions of the limiting behaviour of molecular sieve adsorption columns. Under conditions of micropore diffusion control the concentration dependence of the zeolitic diffusivity causes significant modification of the shape of the constant pattern breakthrough curve for column saturation. For either macropore control or micropore control with a concentration dependent diffusivity, the constant pattern breakthrough curves approach a step function at large values of the non-linearity parameter λ . From the practical point of view this means that, for highly non-linear systems, the breakthrough capacity of the column is almost the same as the saturation capacity and, under these conditions, simple equilibrium considerations are sufficient for design purposes. For such systems the critical factors in process design

will generally be determined by regeneration conditions since, under comparable conditions, regeneration will be much slower than saturation.

NOTATION

a	activity of sorbate
b	Langmuir equilibrium constant
c	sorbate concentration in bulk phase
c_0	sorbate concentration at column inlet
$\bar{c}$	local sorbate concentration in macropore
D_z	zeolitic diffusivity (based on solid area)
D_*	limiting zeolitic diffusivity at zero sorbate concentration
D_p	macropore diffusivity (based on pore sectional area)
D_1	$D_z/D_* = (1 - \lambda\psi)^{-1}$
D'_1	$(1 - \lambda\bar{\psi})^{-2}$
m	ratio of bed void space to zeolite crystal volume = $\epsilon(1 - \epsilon')$
q	local sorbate concentration in a zeolite crystal
$\bar{q}$	average sorbate concentration for a crystal
$\bar{\bar{q}}$	sorbate concentration averaged over a pellet
q^*	sorbate concentration in equilibrium with local sorbate concentration in fluid phase
q_s	saturation sorbate concentration in Langmuir equation
q_0	initial (or final) uniform sorbate concentration in zeolite crystal in equilibrium with fluid phase concentration c_0
Q	fluid flowrate
r	radial coordinate for zeolite crystal

r_z	radius of zeolite crystal
R	radial coordinate for pellet
R_p	pellet radius
s	$\int_0^\psi D_1 d\psi / \int_0^1 D_1 \cdot d\psi$
s'	$\int_0^{\bar{\psi}} D'_1 d\bar{\psi} / \int_0^1 D'_1 \cdot d\bar{\psi}$
t	time
T	dimensionless time = $(D_*/r_z^2)(t - z/v)$
T'	dimensionless time = $\frac{\epsilon_p D_p (t - z/v)}{w(1 - \epsilon_p) R_p^2 b q_s}$
u	$s\eta$
u'	$s'\eta'$
v	linear fluid velocity
V	column volume
w	volume fraction of zeolite crystals to total solid material in a pellet
X	dimensionless distance $\frac{D_* b q_s z}{r_z^2 v m}$
z	distance measured from bed inlet
β	$-\ln(1 - \lambda)$
ϵ	void fraction of bed
ϵ_p	void fraction of pellet
$1 - \epsilon'$	ratio of zeolite crystal volume to total bed volume = $(1 - \epsilon)(1 - \epsilon_p)w$
ϕ	c/c_0
ψ	q/q_0
$\bar{\psi}$	$\bar{q}/q_0$
$\bar{\bar{\psi}}$	$\bar{\bar{q}}/q_0$
λ	q_0/q_s
η	r/r_z
η'	R/R_p
Ω	$3w \frac{(1 - \epsilon_p)}{\epsilon_p} \frac{D_z}{D_p} \left[\frac{R_p}{r_z} \right]^2 \frac{dq^*}{dc}$

Note: Solid phase concentrations are based on unit zeolite crystal volume. The dilution effect of any binder is accounted for in the definition of m

REFERENCES

- [1] VERMEULEN T., *Adv. Chem. Engng* 1958 **2** 147.
- [2] RUTHVEN D. M. and LOUGHLIN K. F., *J. Chem. Soc. Faraday Trans. I*, 1972 **68** 696.
- [3] RUTHVEN D. M. and LOUGHLIN K. F., *Can. J. Chem. Engng*, 1972 **50** 550.
- [4] BARRER R. M. and FENDER B. E. F., *J. phys. Chem. Solids* 1961 **21** 12.
- [5] HABGOOD H. W., *Can. J. Chem.* 1958 **36** 1384.
- [6] RUTHVEN D. M. and LOUGHLIN K. F., *Trans. Faraday Soc.* 1971 **67** 1661.
- [7] DE VAULT D., *J. Am. Chem. Soc.* 1943 **65** 532.
- [8] LIGHTFOOT E. N., SANCHEZ-PALMA R. J. and EDWARDS D. O., *New Chemical Engineering Separation Techniques*, (Ed. SCHOEN H. M.), Interscience, New York 1962.
- [9] GARG D. R. and RUTHVEN D. M., *Chem. Engng Sci.* 1972 **27** 417.
- [10] GARG D. R. Ph.D. Thesis, University of New Brunswick 1972.
- [11] HALL K. R., EAGLETON L. C., ACRIVOS A. and VERMEULEN T., *Ind. Engng Chem. Fundls* 1966 **5** 212.